Exploratory Data Analysis of New York City TLC Data

Executive summary report

Commission Prepared by **Automatidata**

Project Overview

This project focuses on analyzing NYC Taxi trip data to understand ride patterns, driver earnings, and trip costs. By exploring and cleaning the dataset, we lay the groundwork for building a machine learning model that can accurately predict taxi fares. This will help both drivers and the business optimize their daily operations and revenue.

Key Insights

The Problem: Our taxi data contains massive errors, such as negative fares, invalid rate codes, and impossible trip distances, that make it difficult to understand true patterns. If we use this "dirty" data to train a predictive model, the model will make highly inaccurate fare predictions, hurting trust with both drivers and passengers.

Proposed solution: By performing thorough Exploratory Data Analysis (EDA), we clean the dataset by removing illogical records (like negative costs and zero-mile trips).

Keys insights

- The vast majority of taxi rides in NYC are short and highly affordable. In fact, 75% of all trips are under 3 miles long and cost less than $18.00.
- Credit card transactions record tips automatically, whereas cash transactions show $0.00 for tips. This means our tipping data is only reliable for card payments.

Details

As a result of the conducted exploratory data analysis, the Automatidata data team considered trip distance and total amount as key variables to depict a taxi cab ride. The provided scatter plot shows the relationship between the two variables. This scatter plot was created in Tableau to enhance the provided visualization.

Alt Text: Graph displaying New York City TLC data plotting variables for total distance and total amount.

Next Steps

- **Step 1: Clean and Filter the Data:** Explicitly remove negative fare entries, correct invalid rate codes, and filter out extreme statistical outliers.
- **Step 2: Prepare Datetime Features:** Convert pickup and dropoff text fields into actual date/time formats so we can calculate and analyze trip durations.
- **Step 3: Create a Business Dashboard:** Build a simple, interactive visual dashboard to share these key trends and driver earnings insights with business stakeholders.